Soumyadip Kundu

Email: soumyadip23@iisertvm.ac.in **Website:** dipsym.github.io/lampane
INSPIRE-HEP: Author profile

Research Interests

Beyond-the-Standard-Model (BSM) phenomenology; effective field theory; collider phenomenology; statistical techniques such as optimal observables.

Education

Ph.D. in Particle Physics Phenomenology Jul 2023 – Present
School of Physics, IISER Thiruvananthapuram, India
Supervisor: Dr. Tanumoy Mandal

M.Sc. in Physics Sep 2021 – Jun 2023
IISER Thiruvananthapuram, India

B.Sc. in Physics Aug 2018 – Jul 2021
Jadavpur University, Kolkata, India

Publications

Preprints

1. U. Danielsson, R. Enberg, G. Ingelman, **S. Kundu**, T. Mandal, and S. Mitra, "Multi-Higgs production from a theory of varying Higgs quartic and top Yukawa couplings," arXiv:2610.21751 [hep-ph].
2. U. Danielsson, R. Enberg, G. Ingelman, **S. Kundu**, T. Mandal, and S. Mitra, "Testing varying coupling constants through multi-Higgs production at the LHC," arXiv:2606.14303.

Journal Article

1. A. Bhaskar, D. Das, **S. Kundu**, A. A. Madathil, T. Mandal, and S. Mitra, "Vector leptoquark contributions to lepton dipole moments," *Phys. Rev. D* **111**, 015045 (2025), arXiv:2408.11798 [hep-ph].

Conference contribution

1. **S. Kundu**, A. Bhaskar, D. Das, A. A. Madathil, T. Mandal, and S. Mitra, "Exploring TeV-scale vector leptoquark as a solution to lepton dipole moments" (2026).

Teaching and Mentoring

Teaching Assistant, IISER Thiruvananthapuram

- MSP 316: Statistical Mechanics I (Vasanth 2024).
- MSP 314: Quantum Mechanics I (Varsha 2023, 2024; Vasanth 2025).
- Quantum Mechanics II (Varsha 2025).

Minor project mentoring, 2023
IISER Thiruvananthapuram: supervised an M.Sc. first-year student on "Statistical Analysis of Higgs Boson Discovery: Evaluating Significance through the p-value."

Talks and Posters

Poster: Exploring TeV-scale vector leptoquark as a solution to lepton dipole moments Dec 2024
XXVI DAE-BRNS High Energy Physics Symposium, BHU, Varanasi

Poster: Leptoquarks and lepton dipole moments 2025
Frontier Symposium in Physics (FS-PHY 2025)

Conferences and Workshops

Advanced School on Mathematical Methods for Physicists July 2026
IMSc, Chennai

SSHEP 2025: Spring School on High Energy Physics 2025
IMSc, Chennai

DAE-HEP 2024: XXVI DAE-BRNS High Energy Physics Symposium Dec 2024
BHU, Varanasi

GIAN Course: Standard Model Effective Field Theories and Applications to Higgs, Neutrinos, and Dark Matter 2024
IIT Guwahati

PPC 2024: 17th International Conference on Interconnections between Particle Physics and Cosmology 2024
IIT Hyderabad

ML4HEP 2024: Statistical Methods and Machine Learning in High Energy Physics 2024
ICTS, Bengaluru

Awards and Honors

Qualified GATE (Physics) 2023
Qualified JAM (Physics) 2021
DST-INSPIRE Scholarship 2018 – 2023

Skills

Programming and tools: Python, Mathematica, Package-X, MadGraph.
Relevant coursework: Quantum Field Theory I, Particle Physics, General Relativity.
Languages: English, Bengali, Hindi.

Additional Activities

Contributor, PSIt
IISER TVM Physics Society journal
Wrote physics articles in Bengali for science outreach.

Boys Hostel Representative (Mukurthi)
IISER Thiruvananthapuram
Over two years: liaison between students and administration, and organizer of hostel activities.

References

Dr. Tanumoy Mandal, Assistant Professor, IISER Thiruvananthapuram
tanumoy@iisertvm.ac.in

Dr. Subhadip Mitra, Assistant Professor, IIIT Hyderabad
subhadip.mitra@iiit.ac.in

Dr. Shabnam Iyyani, Assistant Professor, IISER Thiruvananthapuram
shabnam@iisertvm.ac.in